CURRICULUM VITAE

Kukhyeon Kim

Ph.D.
61186, 77, Yongbong-ro, Buk-gu, Gwangju
Chonnam National University, South Korea
edu.kukhyeon@gmail.com

UPDATED MARCH 07, 2026

EDUCATION

Postdoctoral Researcher The Institute of Educational Research, Chonnam National University	September 2024 -present
Postdoctoral Researcher BK21 Education Department (Flourishing Education for Future), Chonnam National University	September 2024 - November 2025

EDUCATION

Doctor of Philosophy in Education (Ph.D.) Educational Technology, Chonnam National University	August 2024
Integrated M.S./Ph.D. Student Educational Technology, Chonnam National University	March 2018 - August 2024
English Teacher (Public school) Moon-Jung Girls' Highschool, Gwangju Highschool	March 2016 - February 2019
English Tutor	
Hangil Extracurricular Private Tutoring Academy	March 2012 - February 2013
Home-visiting Tutor	March 2013 - February 2016
Studying for the Public-school teacher examination	March 2012 - February 2016
Bachelor of Education (B.Ed.) College of Education, Chonnam National University	March 2008 - February 2012

ASSISTANTSHIP

BK 21 Scholarship Fellow, Flourishing Education of BK21 Research Group	September 2020 - February 2024
Graduate Research Assistant, The Institute of Educational Research, Chonnam National University	March 2019 – August 2024
Graduate Research Assistant, Center for Immersive Learning Technology (CILT)	March 2018 – August 2024

RESEARCH INTERESTS

AI Digital Competency for Teacher Education
AR/VR/ XR Simulation-based Learning
Learner Experience Design based on Emotion and Cognition
Physiological multimodal data analysis: Eye-tracking, FEA
Virtual Humans and Avatar
Flourishing Education

CERTIFICATES & QUALIFICATIONS

Educational Technology, Chonnam National University Doctor of Philosophy in Education	August 2024

Ministry of Education, Republic of Korea

Secondary School Teacher's Certificate, Pedagogy (Ministry of Education, Republic of Korea)	February 2012
Secondary School Teacher's Certificate, English (Ministry of Education, Republic of Korea)	February 2012

TEACHING EXPERIENCE

Lecturer, Digital Education College of Education, Gwangju National University of Education	March 2026 - Present
Lecturer, Education Technology College of Education, Gwangju National University of Education	March 2026 - Present
Lecturer, Educational Methods and Technology College of Education, Gwangju National University of Education	September 2022 - Present
English Teacher Moon-Jung Girls' Highschool & Gwangju Highschool	April 2016-February 2019

HONORS

Excellence Award for BK21 Participant	Minister of Education	2025
Best Academic Student Paper Award Winners	Immersive Learning Research Network(iLRN) 2024 10th Annual conference	2024
The Grand Graduate Student Award	BK21 Institute of Flourishing Education	2023
First Place Poster	Association for Educational Communications & Technology (AECT) 2023 100th Annual conference	2023
Innovation in Higher Education Award	Immersive Learning Research Network(iLRN) 2023 9th Annual conference	2023
The Grand Graduate Student Award	BK21 Institute of Flourishing Education	2022
Young Scholar Award	International Conference for Media in Education	2022
Showcase Award	Immersive Learning Research Network(iLRN) 2021 7th Annual conference	2021
The Grand Graduate Student Award	BK21 Institute of Flourishing Education	2020
Young Scholar Award	International Conference for Media in Education	2020
The first prize of The Contest for research	The Digital Ethics	2019
Young Scholar Award	International Conference for Media in Education	2019
Grand Prize of Media Festival	The Korean Society for Educational Technology	2019
Young Scholar Award	International Conference of Media in Education	2018
Grand Prize of Media Festival	Spring Conference of the Korean Association for Educational Information and Media	2019

JOURNAL PUBLICATIONS

[Written in English]

Yoon, I., Ryu, J., **Kim, K**., & Kim, I. What AI–Digital Competencies Should Teachers Develop Throughout Their Careers?: Designing a Career-Responsive Framework Through a Delphi study. 11, 1745059*, Frontiers in Education*.

Chae, D., Asami, K., Kim, J., **Kim, K**., Ryu, J., & Doorenbos, A. Z. (2024). Home-based walking intervention for middle-aged migrant women using 360-degree virtual videos and a wearable activity tracker: A mixed-methods pilot study. *Research in Community and Public Health Nursing, 35*(1), 10-21.

Kim, K., Park, S., Ryu, J. & Lim, T. (2023). Scenario Usefulness and Avatar Realism in an Augmented Reality-based Classroom Simulation for Preservice Teacher Training. *Educational Technology International, 24*(1), 1-27

Chae, D., Kim, J., **Kim, K.,** Ryu, J., Asami, K. & Doorenbos, A. (2023). An Immersive Virtual Reality Simulation for Cross-Cultural Communication Skills: Development and Feasibility. *Clinical Simulation in Nursing, 77*, 13-22.

Lee, J., Hollister, J. M., Lim, T., **Kim, K**. & Ryu, J. (2022). A Case Review for the Design of VR-based Training for Enhancing Empathy and Cultural Competency of Public Librarians. *International Journal of Knowledge Content Development & Technology, 12*(Special), 107-134.

Chae, D., Kim, Y., Ryu, J., Asami, K., Kim, J. & **Kim, K.** (2021). E-learning cultural competence for public health workers: A feasibility and pilot study. *Public Health Nursing*, *38*(5), 897-906.

[Written in Korean]

Kim, K., Jeong, J., Jeon, H. & Cha, S. (2025). Analysis of Changes in Middle School Students′ Career Competencies in the Digital Transformation Era. *The Journal of Educational Research, 47*(3), 243-263.

Kim, K. & Ryu, J. (2025). Designing a roadmap for enhancing XR simulation training systems: a Delphi-based approach. *The Journal of Educational Information and Media*, 31(6), 2977-3000

Kang, H., Ryu, J. & **Kim, K.** (2025). The Effect of Extended Reality Simulations for Industrial Safety on User Experience. *Journal of Digital Contents Society, 26*(9), 2513-2525.

Ki, S., **Kim, K**., Han, D., Ryu, J. & Kang, H. (2025). The Effects of Digital Twin-Based Extended Reality Simulations on Task Load, Academic Self-Efficacy, and Learning Achievement. *The Korean Association for Educational Methodology Studies*, 37(3), 297-322.

Yoon, I., Ryu, J. & **Kim, K**. (2025). Validation of a Teacher AI and Digital Competency Scale Based on the Teaching Career Stage Using Exploratory Factor Analysis. *The Korea Educational Review*, 31(2), 95-118.

Kim, K. & Ryu, J. (2024). Design and Development of Extended Reality Simulations for High School Social Studies from a Learning Experience Design Perspective. Journal of Digital Contents Society, 25(7), 1999-2010.

Ryu, J. & **Kim, K**. (2023). The Effect of Presence of Face with Avatar on Social Presence and Attention in Virtual Reality Learning Environment. *The Journal of Educational Information and Media, 29*(4), 1213-1233.

Lim, T., Yang, E., Ki, S**., Kim, K.,** Jeong, Y, Lee, S. & Ryu, J. (2023). ChatGPT-based Keyword Extraction from Text Data: A Comparative Study of Similarity with Manual Methods. *Journal of Educational Technology, 39*(4), 969-1000.

Kim, K., Lee, S. & Ryu, J. (2023). The Effect of Conversational Style with Digital Human on Learning Experience of Emotions and Attention. *The Korea educational review, 29*(3), 27-49.

Kim, K., Yang, E., Ryu, J. & Lim, T. (2023). Use of Facial Expression Analysis and EEG to Analyze Cognitive Load and Emotion of Learners in Virtual Simulation. *The Korean Association for Educational Methodology Studies, 35*(2), 197-221.

Kim, K. & Ryu, J. (2023). Analysis of Life-Like and Bird's Eye Views in the Augmented Reality on User Perceptions, *The Korean Journal of Thinking Development, 19*(1), 1-19.

Kim, K. & Lee, J. (2022). Mediating Effect of Cognitive Self-regulation between Academic Self-efficacy and Flourishing of First-year University Students. *Journal of Learner Centered Curriculum and Instruction, 22*(11), 333-350.

Ryu, J., **Kim, K**. & Yang, E. (2022). The Comparative Study of Attention and Emotion on Parent-Teacher Conference of Online Simulation between In-service and Pre-service Teachers. *The Korea Educational Review, 28*(2), 27-47.

Yang, E., **Kim, K**. & Ryu, J. (2021). A Comparative Study Between In-service and Pre-service Teachers on Scenario Usefulness, Virtual Presence, and Interest Development in Using a Teaching Simulation, *The Journal of Educational Information and Media, 27*(3), 899-923.

Kim, K., Lim, T. & Ryu, J. (2021). A Gaze Behavior Analysis on the Integration Process of Text and Picture in Text Reading. *The Korea Educational Review, 27(*1), 71-89.

Lim, T., Yang, E**., Kim, K**. & Ryu, J. (2021). A study on user experience analysis of high school career education program using metaverse. *Korean Association for Learner-centered Curriculum and Instruction, 21*(15), 679-695.

Kim, C., **Kim, K**., & Ryu, J. (2020). The Effects of Augmented Reality based Interaction and Scenarios on Virtual Presence in Teaching Simulation. *Teacher Education Research,* 59(4), 625-640.

Kim, C., **Kim, K**., & Ryu, J. (2020). The Effect of Experience Level on User Perception in the VR Based Simulation for Communication Training with Virtual Patient. *The Journal of Educational Information and Media, 26*(3), 455-475.

Kim, K., Shin, A., & Ryu, J**.** (2020). The Effects of Computer-Based Teaching Simulation for Pre-service Teachers: a meta-analysis. *Teacher Education Research, 59*(3), 429-444.

Ryu, J. & **Kim, K**. (2020). The Effects of Emotional Interaction with Virtual Student on the User's Eye-fixation and Virtual Presence in the Teaching Simulation. *The Journal of the Korea Contents Association,* 20(2). 581-593.

Ryu, J., **Kim, K**., & Jeong, M. (2019). The Consideration Factors of Ethical Bias for Debate Bot. *The Digital Ethics, 3*(2), 63-76.

Ki, S., **Kim, K**., & Ryu, J. (2019). A Semantic Network Analysis on the Perception of Teaching Competencies of Pre-service Teachers on Teaching Practice. *Teacher Education Research, 58*(4), 699-718.

BOOK CHAPTERS

Ryu, J., Park, S., Yang, E., & **Kim, K**. (2023). Formative Design of Authentic Scenarios for a Virtual Reality-Based Parent-Teacher Conference Training Simulation. In *Formative Design in Learning: Design Thinking, Growth Mindset and Community* (pp. 77-91), Cham: Springer Nature Switzerland.

Park, S., Ryu, J., **Kim, K**., & Kim, C. (2021). Mobile Simulation for Effective Classroom Management Skill Training for Preservice Teachers. In *Learning: Design, Engagement and Definition* (pp. 223-238), Springer.

CONFERENCE PRESENTATIONS (INTERNATIONAL ONLY)

2025

Kim, K., Han, D. & Ryu, J. (2025, October 10-24). Effects of Dual-Task Mental Load on Head Movement, Gaze Behavior, and Behavioral Patterns in XR Simulations [Poster]. *Association for Educational Communications and Technology (AECT)*, Las Vegas, United States.

Ki, S., **Kim, K.**, Han, D. & Ryu, J. (2025, October 10-24). Effects of Digital Twin Simulations on Cognitive Load and Learning in the XR-Based Environment [Paper presentation]. *Association for Educational Communications and Technology (AECT)*, Las Vegas, United States.

Kim, K. & Ryu, J. (2025, October 10-24). The Impact of Extended Reality Simulation Experience on Students' Academic Self-efficacy and achievement [Paper presentation]. *Association for Educational Communications and Technology (AECT)*, Las Vegas, United States.

Ryu, J., **Kim, K**., Yang, E., Son, U., Lee, J., & Lee, B. (2025, October 10-24). XR-LEADS: A Multimodal, Data-Driven, Real-Time Learning Analytics Dashboard for Designing XR-Based Learning Experiences [Paper presentation]. *Association for Educational Communications and Technology (AECT)*, Las Vegas, United States.

Kim, K., Han, D., & Ryu, J. (2025, August 7-9). Understanding Pre-service Teachers' Learning Community Experiences from the Perspective of Flourishing [Roundtable]. *International Conference for Media in Education (ICoME),* Incheon, Korea.

Kim, K., & Ryu, J. (2025, June 17–18). Development of XR Simulation for Geography Learning based on LXD [Paper presentation]. *The Immersive Learning Research Network (iLRN)*, Chicago, United States.

Kim, K. (2025, May 9-11). Exploring Teachers' Competencies in AI-Convergence Educational Environment [Oral Presentation]. *Asian Students' SEminar and Round Table (ASSERT)*, Gwangju, South Korea.

Kim, K., & Ryu, J. (2025, April 22–27). The effect of XR-based social studies simulation on learning experience [Paper presentation]. *American Educational Research Association Annual Meeting (AERA),* Denver, United States.

Ryu, J., **Kim, K**., Yang, E., Son, U., Lee, J., & Lee, B. (2025, March 17–21). XR-LEADS: A multimodal framework for interactive learning experience and real-time analytics in XR training [Paper presentation]. *Society for Information Technology and Teacher Education Conference (SITE),* Florida, United States.

Kim, K., Lee, S., & Ryu, J. (2025, March 17–21). The analysis on AI-convergence education competencies of teachers [Paper presentation]. *Society for Information Technology and Teacher Education Conference (SITE),* Florida, United States.

Ryu, J., **Kim, K**., Yang, E., Son, U., Lee, J., & Lee, B. (2025, March 17–21). XR-LEADS: AI-powered real-time learning analytics for XR training [Demo presentation]. *Society for Information Technology and Teacher Education Conference* (SITE), Florida, United States.

2024

Yang, E., **Kim, K**., Jeong, Y., Kim, D., Kang, H., Lee, S., & Ryu, J. (2024, October 19–23). Designing extended reality-based training content for cardiopulmonary resuscitation skills for laypersons [Showcase]. *Association for Educational Communications & Technology International Convention (AECT)*, Kansas City, Missouri, US.

Kim, K., & Ryu, J. (2024, October 19–23). The relation of learners' cognitive, affective, psycho-motor, and learning experience in XR-based geographic learning simulation [Concurrent session]. *Association for Educational Communications & Technology International Convention (AECT)*, Kansas City, Missouri, United States.

Kim, K., & Ryu, J. (2024, August 22). The Effect of XR-based Geography Learning on Academic Self-efficacy and Achievement [Concurrent]. *International Conference for Media in Education (ICoME),* Tokyo, Japan.

Jeong, Y., **Kim, K**., & Ryu, J. (2024, June 5). The Impact of Cognitive, Affective, and Psychomotor Learning Perception on Learning Outcomes in the Extended Reality based Nursing Simulation [Full Paper]. *The Immersive Learning Research Network (iLRN)*, online.

Jeong Y., Yang E., **Kim, K**., Kim, D. & Ryu, J. (2024, May 18-22). Machine Learning Insight into Geography Education Enhancement in XR Environments. *9th International XR-Metaverse Conference (IAITI)*, Busan, South Korea.

Kim, D., Yang E., **Kim, K**., Jeong T., & Ryu, J. (2024, May 18-22). Developing Extended Reality Simulation for Core Basic Nursing Skills. *International XR-Metaverse Conference (IAITI),* Busan, South Korea.

Kim, K., Park, S., Ryu, J., & Lim, T. (2024, April 11-14). Augmented Reality-Based Classroom Management Simulation. *American Educational Research Association (AERA),* Philadelphia, United States.

Kim, K., Jeong, Y., & Ryu, J. (2024, March 25-29). Effect of Visual Supporting Aid on Cognitive Load in VR-based Parent-Teacher Conference Simulation. *Society for Information Technology and Teacher Education (SITE)*, Las Vegas, United States.

2023

Kim, K., Cheon, J., Robert Davis, & Ryu, J. (2023, October 15-19). The Design and Development of AR-based Historical Event Learning Application and Students' Satisfaction [Poster]. *Association for Educational Communications & Technology International Convention (AECT)*, Orlando, United States.

Kim, K., Yang, E., & Ryu, J. (2023, October 15-19). Design, The effects of the interactive virtual human's role in conversation on the user's persona [Poster]. *Association for Educational Communications & Technology International Convention (AECT)*, Orlando, United States.

Kim, K. & Ryu, J. (2023, August 16-18). Effect of Verbal Interaction with Virtual Human on Emotion and Attention [Round table]. *International Conference for Media in Education (ICoME)*, Wenzhou, China.

Kim, K., Park, S., & Ryu, J. (2023, June 27-30). The Effect of Conversation Theme on User's Valence toward Virtual Human [session presentation]. *Positive Technology International Conference*, Hong Kong.

Kim, K., Kim, J., Kim, D., Xiang, H., Park, S. & Ryu, J. (2023, May 18-20). Multimodal Data as Users Performance Recording in XR-based Training Simulation Environment [paper session]. *The Immersive Learning Research Network (iLRN)*, Virtual Campus.

Ryu, J., Park, S. Yang, E., & **Kim, K.** (2023, April 13-16). Authentic Scenario Design for Virtual Reality-based Parent-Teacher Conference Training Simulation [paper session]. *American Educational Research Association (AERA)*, Chicago, United States.

2022

Kim, K., Xiang, H., & Ryu, J. (2022, August). The Exploratory Analysis of Spatial Perception in Augmented Reality with Gaze path and Task Achievements. *International Conference for Media in Education (ICoME)*, Metaverse Conference.

Chae, D., Kim, J., **Kim, K.,** Ryu, J., & Asami, K. (2022, July). Virtual Reality Simulation to Enhance the Cultural Competence of Nursing Personnel: A Feasibility Study. *Sigma's 33rd International Nursing Research Congress*.

Park, S., Ryu, J., Yang, E. & **Kim, K**. (2022, July 7-8). Formative Design of Authentic Scenarios for Virtual Reality-based Parent-teacher Conference Training Simulation [round table]. *Association for Educational Communications & Technology International Convention* Summer Research Symposium (AECT SRS), online.

Kim, K., Yang, E., & Ryu, J. (2022, May). Work-in-progress—The Effect of Students' Perceptions on Intention to use Metaverse Learning Environment in Higher Education. *The Immersive Learning Research Network (iLRN)*, online.

Xiang, H., **Kim, K.,** & Ryu, J. (2022, May). Work-in-Progress—Preliminary Analysis of Spatial Perception in AR application with Eye-Tracking Data. *The Immersive Learning Research Network (iLRN)*, online.

Yang, E., **Kim, K**., & Ryu, J. (2022, April). SimTEACHER: VR-based Simulation for Pre-Service Teachers. *Society for Information Technology and Teacher Education Conference (SITE)*, online.

Kim, K., Jeong, Y., & Ryu, J. (2022, April). Does the Real Face Provision Improve the Attention and Social Presence in Metaverse as Learning Environments. *Society for Information Technology and Teacher Education Conference (SITE),* online.

2021

Kim, K., Yang, E., & Ryu, J. (2021, November). Effects of counseling interaction of In-service teacher in parents-teacher conferencing simulation on facial expression and scenario usefulness [Poster session]. *Association for Educational Communications & Technology International Convention (AECT)*, online.

Kim, K., Jo, S. & Ryu, J. (2021, October). The Scenario Components of Teaching Simulations for Classroom Management. *Society for Information Technology and Teacher Education Conference Interactive (SITE),* online.

Kim, K. & Ryu, J. (2021, August). Emotional Reaction Comparison of In-Service Teacher's Gender in Virtual Parent Meeting Simulation [Round table]. *International Conference for Media in Education (ICoME)*, Metaverse Conference.

Ryu, J., Chae, D., **Kim, K**., Kim, J., Asami, K., Jo, S., Xiang, H., & Kim, D. (2021, May). The Development of Virtual Reality Simulation for the Culture Competence in Nursing education. *The Immersive Learning Research Network (iLRN)*, online.

2020

Park, S., Ryu, J., Kim, C., & **Kim, K**. (2020, November). Design and Validation of Culturally Contextualized Classroom Management Scenarios for Virtual Teaching Simulation. *Association for Educational Communications & Technology International Convention (AECT)*, online.

Kim, K., Kim, C., Park, S., Ryu, J., & Lee, S. (2020, November). SimTEACHER Mobile: Design and Development of Immersive Classroom Management Simulation for Pre-service Teacher Training. *Association for Educational Communications & Technology International Convention (AECT)*, Online.

Shin, A. & **Kim, K**. (2020, August). The effects of Computer-based Teaching Simulation for Pre-service Teachers: a meta-analysis [Round table]. *International Conference for Media in Education (ICoME)*, online.

Kim, K., Kim, C., Kim, D., & Ryu, J. (2020, August). SimTEACHER Mobile: Design and Development of Augmented Reality Simulation for Preservice Teacher? The Classroom Management Skill Training. *International Conference for Media in Education (ICoME)*, online.

Kim, K., Kim, C., Cho, M., & Ryu, J. (2020, August). The automatic evaluation model for argumentative interaction in debating chat-bot program. *International Conference for Media in Education (ICoME)*, online.

Park, S., Ryu, J., Kim, K., & Kim, C. (2020, July). Design and Development of Mobile Simulation for Preservice Teacher's Classroom Management Skill Training. *Association for Educational Communications & Technology International Convention Summer Research Symposium (AECT SRS)*, online.

Park, S., Ryu, J., Kim, K., & Kim, C. (2020, June). SimTEACHER Mobile: Augmented Reality-Based Mobile Simulation for Preservice Teacher Classroom Management Skill Training. *The Immersive Learning Research Network (iLRN)*, online.

2018-2019

Ryu, J., Kim, K., & Kim, C. (2019, October). Effects of Auditory Information and Task Difficulty on User's Presence in Haptic-Based Simulation. *Association for Educational Communications & Technology International Convention (AECT)*, Las Vegas, United States.

Kim, C., Ryu, J., & Kim, K. (2019, October). Development of the Virtual Reality Simulation Using Problem Avatar Student for Pre-Service Teachers. *Association for Educational Communications & Technology International Convention (AECT)*, Las Vegas, United States.

Kim, K., Ryu, J., Cho, M., & Lee, Y. (2019, August). UX Design of Augmented Reality for the Elementary School Students. *International Conference for Media in Education (ICoME)*, Shenzhen, China.

Ryu, J., Kim, K., & Kim, C. (2019, March). The effect on the scenario level on the eye-fixation time on the virtual problem student avatar in virtual classroom simulation. *Society for Information Technology and Teacher Education Conference (SITE)*, AACE, Las Vegas, United States.

Kim, K., & Kim, S. (2018, August). Exploring Nonverbal Behavior interactions of Virtual Problem Student Avatar Using Eye-Tracking in Virtual Teaching Simulation [Poster Session]. *International Conference for Media in Education (ICoME)*, Cheongju, Korea.